Compiled Problems – Senior Division

1. **01-02 C1 Computer Number Systems**

   Solve for X_8: $BAD_{16} = X_8 + 465_8 - 447_8$

2. **01-02 C1 Computer Number Systems**

   How many 1's are in the binary representation of the following hexadecimal sum?

   $$59A_{16} + B28D6_{16} + E3E_{16}$$

3. **01-02 C2 Computer Number Systems**

   Solve for X_2: $X_2 = F123_{16} - 123_{10} - 123_8$

4. **02-03 C1 Computer Number Systems**

   What hexadecimal number is the next element of the given sequence?

   $$1_{10},\ 11_2,\ 11_8,\ 33_8,\ 1010001_2,\ 363_8,\ _____{}_{16}$$

5. **02-03 C1 Computer Number Systems**

   What is the smallest 3-digit hexadecimal number that when added to $92B_{16}$ will make the sum a hexadecimal palindrome (reads the same from both directions, ex. F7F.

6. **02-03 C2 Computer Number Systems**

   What is the hexadecimal value of $(FF_{16})^2$?

7. **03-04 C1 Computer Number Systems**

   Evaluate each expression and convert the answer to binary.
   Ignoring leading zeros, which binary answers contain more 1's than 0's?
   List all of them.

   A. $431_8 - 11_{16}$ B. $1493_{10} + 72_8$ C. $4B7_{16} + 32_8$
   D. $101_2 + 101_8$ E. $B21_{16} - 72_{10}$

8. **03-04 C1 Computer Number Systems**

   What is the remainder in octal when CA_{16} is divided by AC_{16}?

9. 03-04 C2 Computer Number Systems

Solve for X_{10}: $4_{10} * 1011_2 + 3_{10} * 126_8 - 2_{10} * DF_{16} = X_{10}$

10. 04-05 C1 Computer Number Systems

Solve for X_{16}: $2 * X_{16} + 11011010_2 = 4320_8$

11. 04-05 C1 Computer Number Systems

Which expression has the most 1's in its binary representation:

A) 2345_8 B) FAB_{16} C) 511_{10}
D) 1011011101111_2 E) 3794_{16}

12. 04-05 C2 Computer Number Systems

Solve for X_2: $31_8 * X_2 + 12_{16} = 28_{16} * X_2 - 1001000_2$

13. 05-06 C1 Computer Number Systems

Solve for X_{16}: $(X_{16})^2 - 1010101_2 * X_{16} + 1356_8 = 0$

14. 05-06 C1 Computer Number Systems

Solve for X_2: $504_8 X_2 - 2E6_{16} = 80_{10} X_2 + 11101010_2$

15. 05-06 C2 Computer Number Systems

Find the mean (average) of the following numbers. Express the answer in binary.

10010_2, 20_{16}, 31_8, 11001_2, 25_{10} and 19_{16}

16. 06-07 C1 Computer Number Systems

Simplify: $F_{16} + FF_{16} + \ldots + FFFFFF_{16}$

17. 06-07 C1 Computer Number Systems

Solve for X_2: $24_8 * X_2 - AC_{16} = 101_2 * X_2 - 235_8$

18. 06-07 C2 Computer Number Systems

Solve for X_{16}: $367_8 + 10111_2 = X_{16} - A12_{16}$

19. 07-08 C1 Computer Number Systems

Evaluate and express the answer in hex: $101_2 * 32_8 + 15_{10} (24_{16} - 111_2)$

20. 07-08 C1 Computer Number Systems

Solve for X: $1X6_{16} = X26_8$

21. 07-08 C2 Computer Number Systems

How many full 3-bit octal numbers are palindromes and are perfect squares in base 10? (A palindrome is the same forward and backward.)

22. 08-09 C1 Computer Number Systems

Find the volume in hexadecimal units of a box with dimensions:
$L = 31_8$, $W = 10110_2$, $H = 3F_{16}$

23. 08-09 C1 Computer Number Systems

Solve for X: $10_2 * X_{16} + 2AE_{16} = 4710_8$

24. 08-09 C2 Computer Number Systems

Which of the following have the same value? List the letter(s).

A. 292_{10} B. $113_{16} + 11_{16}$ C. 100100110_2 D. $513_8 - 46_8$

25. 09-10 C1 Computer Number Systems

Solve for X_{16}: $22_8 * X_{16} - 1011101_2 = 20_{16} + 67_8$

26. 09-10 C1 Computer Number Systems

Evaluate the following: $X_{16} = FEED_{16} + DEAF_{16} - BEAD_{16} + FACE_{16} - BEEF_{16}$

27. 09-10 C2 Computer Number Systems

Given a hexadecimal number, add it to its digits reversed. If the resulting sum is a not a palindrome (same read forwards and backwards), then repeat the process until a palindrome is achieved. What is the palindrome? The starting hexadecimal number is BE.

28. 10-11 C1 Computer Number Systems

Which hexadecimal number has the most 5's in its octal representation?

$ACED_{16}$ $BEAD_{16}$ $CAFE_{16}$ $DEED_{16}$ $FACE_{16}$

29. 10-11 C1 Computer Number Systems

How many numbers are palindromes in the hexadecimal representations for the numbers 21_8 to 424_8?

30. 10-11 C2 Computer Number Systems

A crazy pirate buried his treasure on an island and left these directions. He only travels N or S. From the starting point take 1010_2 paces N, 74_8 S, $5A_{16}$ N, 1100100_2 S, 24_8 S, 32_{16} N, 1010000_2 N. How far from the starting point was the treasure buried? Express paces in hexadecimal digits and indicate N or S.

31. 11-12 C1 Computer Number Systems

How many 1's appear in the binary representation of the hexadecimal numbers from 24_{16} to 36_{16} inclusive?

32. 11-12 C1 Computer Number Systems

Find the number that satisfies the following: $143_X = 11*X + 11$

33. 12-13 C1 Computer Number Systems

Find the mean (average) of the following numbers and express it in octal.

170_8 144_{10} 1110000_2 97_{16} 200_8

34. 12-13 C1 Computer Number Systems

Solve for X_{16}: $1011011_2 + 2 * X_{16} = 7321_8$

35. 13-14 C1 Computer Number Systems

Solve for X_{16}: $2_{10} * X_{16} - 464_8 = 11101110_2 - X_{16}$

36. 13-14 C1 Computer Number Systems

What is the next integral decimal number after 328_{10} to have the substring 0110 appear in its binary representation?

37. 14-15 C1 Computer Number Systems

Which is the largest?

a) $11001010_2 + 13_8$
b) $257_8 + 24_{16}$
c) $B4_{16} + 11110_2$
d) $128_{10} + 23_8 + 101010_2 + 17_{16}$
e) $136_{10} + 1001110_2$

38. 14-15 C1 Computer Number Systems

Evaluate the following expression and express the answer in octal:

$$2_{10} * 416_8 + A23_{16} - 10110_2 + 31_8 / 101_2$$

39. 15-16 C1 Computer Number Systems

What is the smallest integral value that can be used for X?

$$15D_{16} < X_{10} + 1010_2 < 1271_8$$

40. 15-16 C1 Computer Number Systems

Find the next term in the following sequence and express the answer in hexadecimal.

$$1_2, \quad 100_2, \quad 20_8, \quad 100_8, \quad 100_{16}$$